

Assignment 2 game design document

Game Name :run for life

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My game Run for Life is inspired by a game called Jetpack Joyride, an old game that I liked when I was a child. I will simplify and change the implementation of the game . In the Jetpack Joyride it is a vertical and horizontal movement, but my game will be 2D from an upper angle so it is left, right, front and back implementation .

Backstory:

- You went to a haunted house for adventure but you awaken the ghost , now the ghost is chasing you and you have to run to get out .

Characters:

- player , the user will control it to escape the haunted house;

Gameplay Mechanics:

- player will need to keep running , if player stops and cannot catch up with the screen or hit a bomb more than the health they have , they died.
- To win, players must reach the targeted meters to win the game.
- game speed goes faster and faster , bomb will spawn more often by time.
- Use wasd for control, D is forward, S is left, W is right and A is backward.
- player have 3 hearts .
- The speed of the bomb will go faster and faster , same with player movement speed, and the bomb spawning rate.

Detail Game Flow:

- When the player starts the game, the bomb will drop from the rightest of the screen , and it will move towards the left of the map. The player will need to avoid touching the bomb . If they touch , they will lose 1 health.
- They have to constantly move forward , if they touch the leftest edge of the screen , they died.
- If they reach the goal meter , they will see the goal presented as a door,when they reach it they win the game .
- speed(the force pulling player back , bomb moving speed , player left right up down speed) will increase by time , bomb spawn rate will keep increasing by time.

Technical Details:

- In the game, players and bomb will be given x,y and speed for display ,updating location and movement.
- All objects will have a constant backward speed, it is applied to bombs and players , the bomb can only move from right to left, players can move freely with wasd control. Both bomb and player will have a constant force that will pull them from right to left. If a player wants to fight against the force . They will have to keep moving forward to counter it.
- There is a definition of how meter is calculated.
- bomb will not overlap .

- Check the sketch of bomb spawning , even when it reaches the max difficulty , there will always be an empty spot that bomb will not spawn.
- At the beginning , bombs will spawn slower , for example spawn every 10 frames. When it is closer to the end, a bomb will spawn more frequently eg 2 frame .
- There will be visual feedback when a player hits a bomb. It will keep flashing white light for a period after the hit.
- When the player presses the setting button, it will stop the game.
- Canvas is used for the graphic of the game .
- HTML,JavaScript is used.
- The ground has animation, if will keep scrolling from right to left for simulation if goal is not reached.

Math details:

- There is a variable called difficultyIncreaseRate, it will control the speed of the game , player change of speed and the spawn rate of the bomb, the high this value is , the harder the game becomes.
- At first difficultyIncreaseRate will be 0.05 . Every 250 frame difficultyIncreaseRate will increase by 0.05
- The other variable is call pulledSpeed . Player and bomb will have this force apply on them constantly. A player with no control or bomb will automatically move to the left.
- pulledSpeed is setted to 0.5.
- bomb spawn rate equation is as follows: $\text{spawnBombByFrame} = \frac{250}{(1+\text{difficultyIncreaseRate})}$.
- bomb movement speed : $\text{pulledSpeed} + \text{difficultyIncreaseRate}$, and this will be the speed the bomb moves to left
- player movement will also be affected by the pulledSpeed and difficultyIncreaseRate ,player base speed is 1 . The equation is : $(\text{player speed} + \text{difficultyIncreaseRate}) - (\text{pulledSpeed} - \text{difficultyIncreaseRate})$.The player base speed will apply only when keyboard input is pressed (wasd) ,otherwise player speed is 0 and the player will be pulled to the left.
- For meter calculation every 1 unit in the game is 0.1 meter.
- Every frame , there will be hit box check, if the player hits the bomb, player HP will reduce by 1 .
- After the player hits the bomb , it will trigger the Invincibility window , in this period , player will not get hurt again.
- goal meter is randomly set between 100 - 1000 meters.
- the ground animation have an equation , when goal is not reached , $\text{backgroundX} -= \text{pulledSpeed} + \text{difficultyIncreaseRate}$, else $\text{backgroundX} = 0$

System file(possible file):

- UI layer:
 - menu
 - setting screen(resume or restart(go to main menu))
 - show bombs
 - show players
 - show goal
 - show ghost
- Logic layer:

- position and speed system
- pull back system
- meter counting system
- hit box
- bomb spawn
- win lose check
- HP system
- invincible effect
- difficulty system

System file(actual file):

- UI layer:
 - index.html
- UI and logic layer:
 - Game.js
- logic layer:
 - Button.js
 - GameHelper.js
 - Main.js
 - Meter.js
- resources (png)

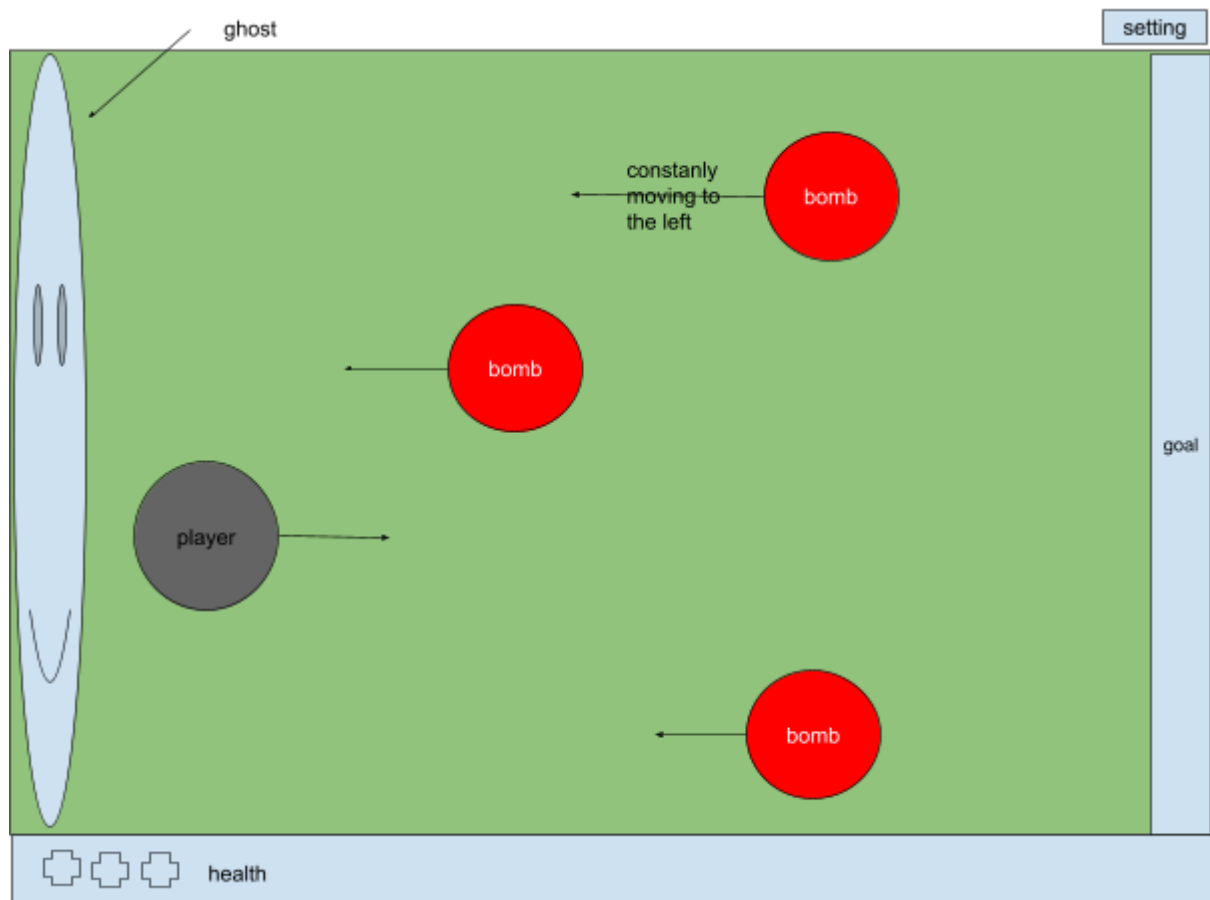
future goal :

- on the map it will spawn some special item like heals , speed up , spawn less bomb , etc.
- ranking system
- endless mode
- more level
- music or sound effect

in game picture:



idea sketch:



use random generator to pick the
locaiton

